

Mini Project

Electrical Design Office Management System
EE4207 Web Application Development

A Report Submitted to the
Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

On 31st October 2026

EG/2023/5734 - Menike E.P.R.R.M.R.N.B.E.

1. Introduction

Preparing for technical interviews can be challenging for students and recent graduates. Candidates need to practise interview questions, assess their knowledge, track their progress and learn from the experiences of others. Managing these activities using separate resources can make preparation less organised.

The Interview Preparation Portal was developed as a web application to support interview preparation through a single platform. It provides question practice, quizzes, progress information, bookmarks and a section for sharing interview experiences.

The project was developed as an individual mini project for EE4207 – Web Application Development. It applies web development concepts such as frontend design, backend API development, database integration, user authentication and input validation.

1.1 Problem Statement

Students preparing for technical interviews may find it difficult to organise practice questions, evaluate their performance and keep useful preparation materials in one place. They may also have limited access to practical interview experiences shared by other candidates.

Therefore, a centralised web application is useful for organising interview preparation activities and supporting continuous practice.

1.2 Objectives

The main objectives of this project are:

- To develop a web-based interview preparation platform.
- To provide a question bank for technical practice.
- To allow users to attempt quizzes and view their results.
- To provide a dashboard for viewing preparation progress.
- To allow users to bookmark useful questions.
- To support sharing and managing interview experiences.
- To apply database operations, input validation and authentication in a web application.

1.3 Scope

The application focuses on technical interview preparation. Its main functions include user login, a question bank, quizzes, performance information, bookmarks and interview experience management.

The application is intended to provide a practical preparation environment. It does not guarantee interview success, and the usefulness of the platform depends on the quality and coverage of its questions and shared experiences.

2. System Overview and Technologies

The Interview Preparation Portal uses a client-server architecture. The frontend provides the user interface, the backend processes requests and applies application logic, and the database stores application data.

2.1 Technologies Used

React: Used to develop the interactive user interface and reusable interface components (React, n.d.).

Vite: Used as the frontend development and build tool (Vite, n.d.).

Node.js: Provides the JavaScript runtime for the backend (Node.js, n.d.).

Express.js: Used to create backend routes and handle HTTP requests (Express.js, n.d.).

MongoDB Atlas: Used as the cloud-hosted database for storing application data (MongoDB, n.d.).

JavaScript: Used for frontend and backend application logic.

HTML and CSS: Used to structure and style the user interface.

Git and GitHub: Used for version control and storing the project source code in a repository (Git, n.d.).

2.2 System Architecture

The system follows a client-server architecture. The user interacts with the React frontend, which sends requests to the Express backend. The backend processes the requests and communicates with MongoDB Atlas when data needs to be retrieved or stored. The response is then returned to the frontend and displayed to the user.

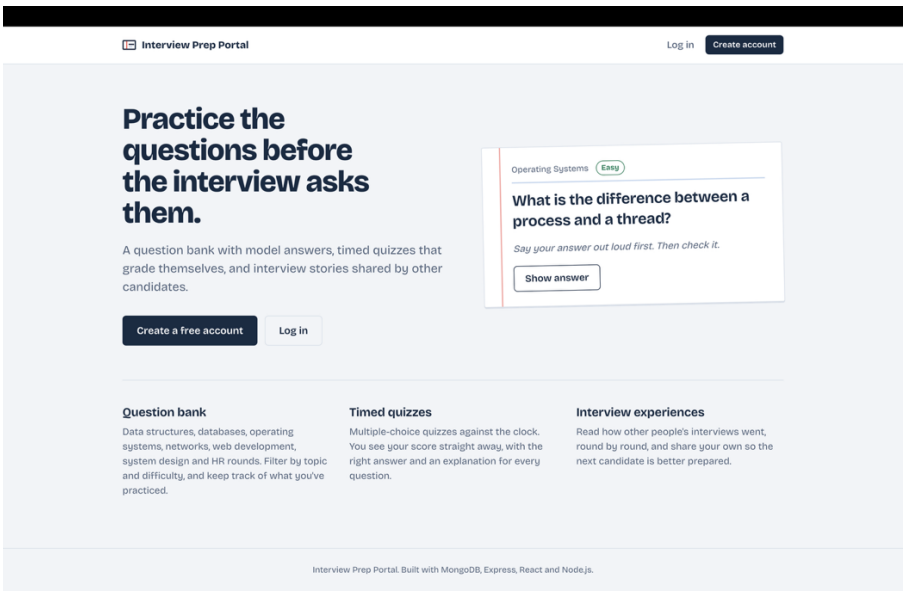


Figure 1: Architecture of the Interview Preparation Portal

3. Database Design and Implementation

MongoDB Atlas is used to store the application's data. A document-based database structure was selected to support the application's different data types.

The main data models used in the project are described below. These models support the storage and management of information used by the application.

3.1 User Model

The User model stores information required to identify and manage user accounts. It supports authentication and the association of user-specific activities with the relevant account. Access restrictions are applied to relevant operations where required.

3.2 Question Model

The Question model stores interview practice questions and their related information, such as question content, topic, difficulty and answer-related data where implemented. These records form the question bank used for practice.

3.3 Quiz Model

The Quiz model stores information associated with quizzes, including the questions or quiz configuration used to organise an assessment.

3.4 Attempt Model

The Attempt model records quiz attempts and associated results. This information can be used to display previous performance and support progress tracking.

3.5 Experience Model

The Experience model stores interview experiences shared by users. The information includes details such as company, role, number of rounds, outcome, difficulty and descriptions of interview rounds, depending on the submitted form.

These models support the main operations of the application, including retrieving information, creating records and updating or deleting permitted records.

4. Application Features

4.1 User Authentication

The application provides a login interface through which users can access the portal. Authentication requests are handled by the backend. Access to protected functionality is controlled according to the application's authentication and authorisation logic.

4.2 Dashboard

The dashboard provides an overview of the user's preparation activities. It displays available progress information and links to the main sections of the portal.

The dashboard helps users access practice activities and review their preparation progress from one location.

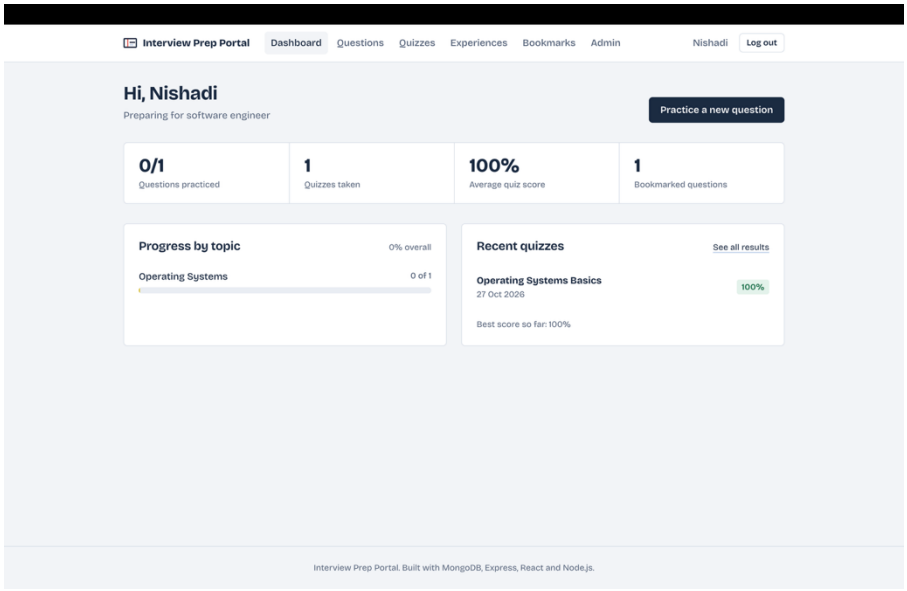


Figure 2: Dashboard of the Interview Preparation Portal

4.3 Question Bank

The Question Bank allows users to access interview practice questions. Questions can be organised using information such as topic and difficulty.

The application also supports question management operations where permitted. Input validation is used to help prevent incomplete or invalid question submissions.

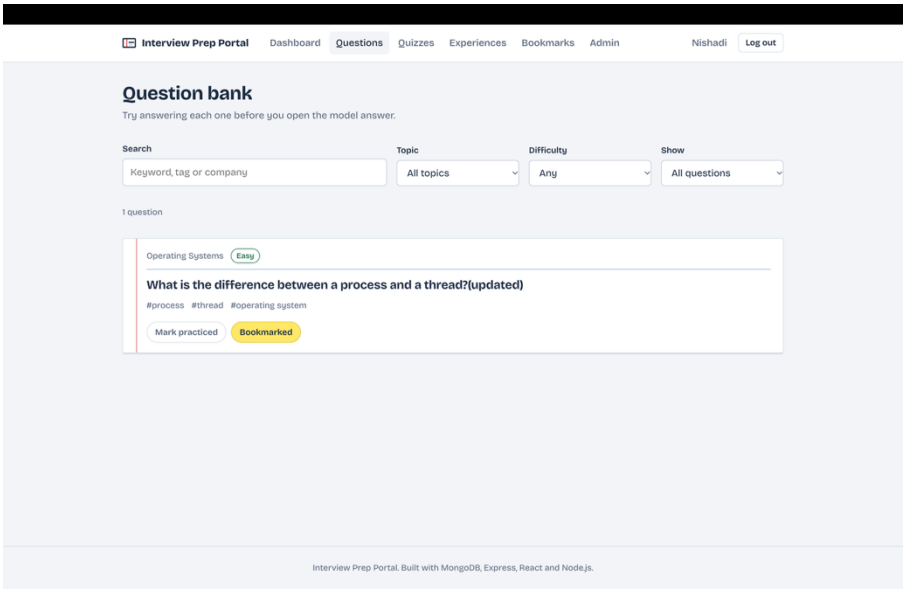


Figure 3: Question Bank interface.

4.4 Bookmarks

The bookmarking feature allows users to save useful questions for later review. This helps users return to selected preparation materials without having to locate them again in the full question bank.

4.5 Quiz and Assessment

The quiz feature allows users to answer questions and evaluate their knowledge. After completing a quiz, the application displays the result.

This feature supports self-assessment and helps users identify areas that may require further practice.

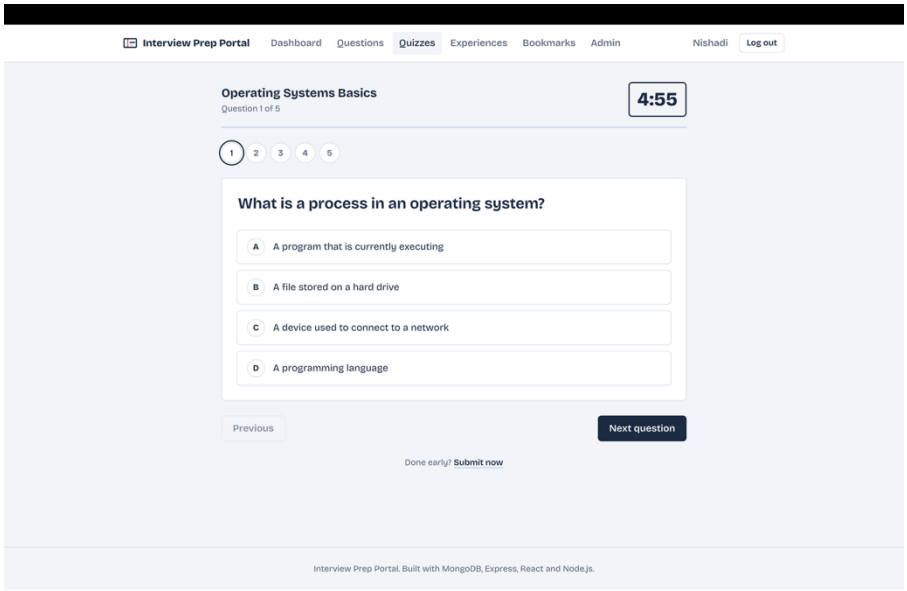


Figure 4: Quiz interface.

4.6 Results and Progress Tracking

The results page displays the outcome of a completed quiz. Quiz attempt information can also support the display of previous results and preparation progress.

Together with the dashboard, this feature helps users review their performance and plan further practice.

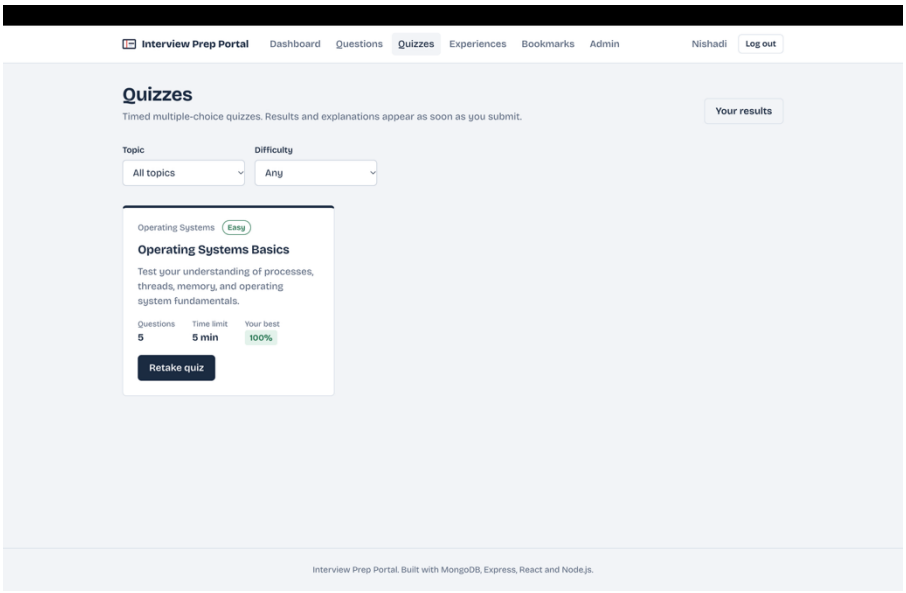


Figure 5: Quiz results interface.

4.7 Interview Experiences

The Interview Experiences section allows users to share information about interviews they have attended. The form includes fields for company, role, location where applicable, number of rounds, outcome, difficulty and descriptions of interview rounds.

Users can view submitted experiences and manage their own posts through the available editing and deletion functions. Required-field validation helps ensure that essential information is entered before submission.

This feature provides a way for candidates to exchange practical information that may help others prepare for interviews.

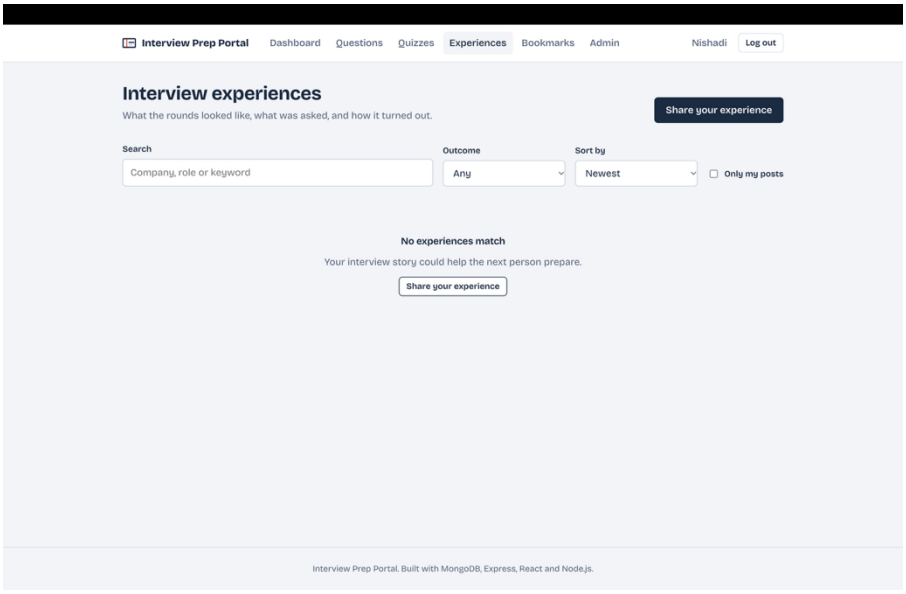


Figure 6: Interview Experiences page.

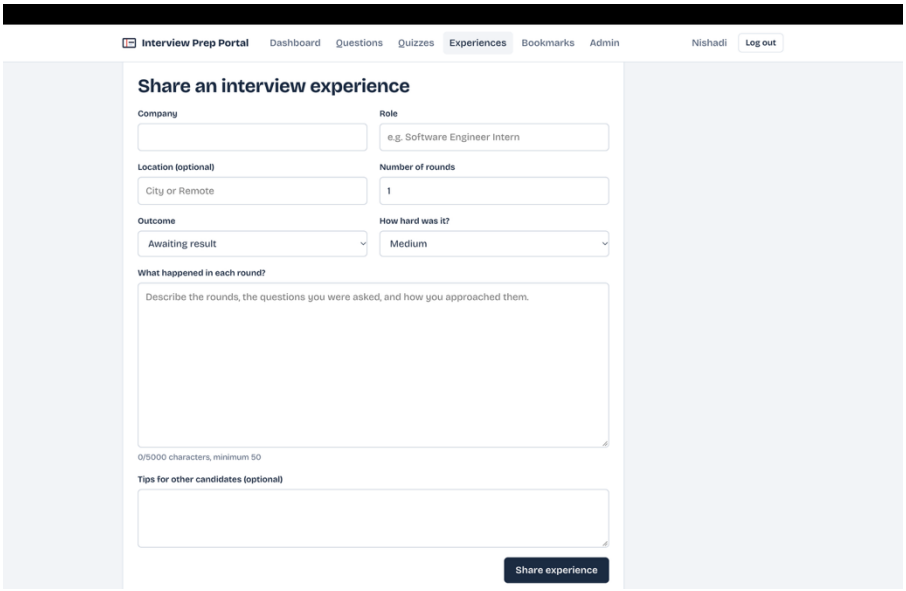


Figure 7: Form for submitting an interview experience.

5. Testing and Validation

The application was checked through practical use of its main interfaces and selected operations. The purpose of this testing was to confirm that important features responded as expected and that basic input validation was working.

5.1 Functional Testing

The following table summarises the checks performed during development.

Test area	Test performed	Observed result
Backend and database	Started the backend and checked its database connection	Connection was established after the network access configuration was corrected
User login	Submitted login details	Login succeeded with the available test account
Question Bank	Opened the question bank	The page displayed the question bank interface
Quiz	Opened and attempted a quiz	Quiz interaction and results were checked
Interview Experiences	Created a test experience	The submitted experience appeared in the listing
Interview Experiences	Edited the test experience	The updated information was displayed
Interview Experiences	Deleted the test experience	The test post was removed
Input validation	Submitted the experience form without required information	The form showed a required-field validation message

Table 1: Functional Testing and Validation Results

5.2 Input Validation

Input validation is important for improving data quality and helping users understand what information is required. During testing, submitting the interview experience form without filling in required fields produced a validation message.

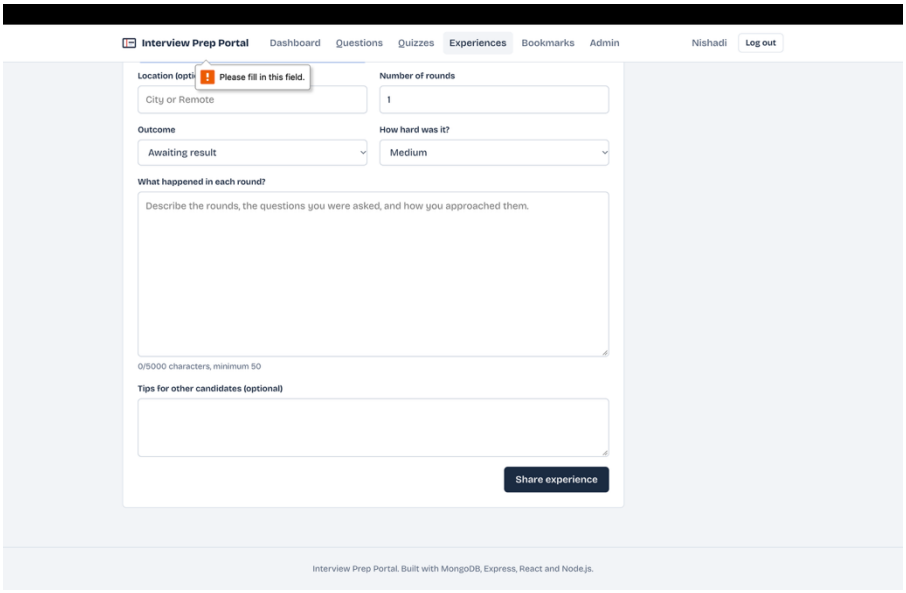


Figure 8: Validation message displayed for incomplete input.

5.3 Testing Limitations

The checks described above cover selected application workflows. They should not be interpreted as exhaustive testing of every possible input, user permission, security condition or failure scenario. Further testing would be useful before deploying the application for general public use

6. Limitations and Future Improvements

Although the application provides several useful interview preparation functions, there are areas that could be improved in future versions.

Possible improvements include:

- Expanding the question bank to cover more technical subjects and difficulty levels.
- Adding more detailed analytics to identify strengths and weaknesses.
- Improving search and filtering for questions and interview experiences.
- Introducing automated testing for important frontend and backend workflows.
- Improving security checks and error handling before public deployment.
- Deploying the application to a production hosting environment.
- Improving the moderation and quality control of user-submitted interview experiences.

These improvements could make the platform more comprehensive, reliable and useful for a wider range of candidates.

7. Conclusion

The Interview Preparation Portal was developed as a web application to support technical interview preparation in one place. It brings together a question bank, quizzes, results, dashboard information, bookmarks and interview experience management.

The project provided practical experience in developing a React frontend, creating backend services using Node.js and Express, connecting to MongoDB Atlas, implementing application features and validating user input.

The main objectives were addressed through the development of the core application features and the testing of selected workflows. Further testing, content expansion and deployment improvements would help strengthen the application in future versions.

References

1. React. (n.d.). *React Documentation*. https://react.dev/
2. Vite. (n.d.). *Vite Documentation*. https://vite.dev/
3. Node.js. (n.d.). *Node.js Documentation*. https://nodejs.org/docs/
4. Express.js. (n.d.). *Express Documentation*. https://expressjs.com/
5. MongoDB. (n.d.). *MongoDB Documentation*. https://www.mongodb.com/docs/
6. Git. (n.d.). *Git Documentation*. https://git-scm.com/doc

Accessed: 30/10/2026